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Performance analysis of single-phase immersion cooling system of data center using FC-40 dielectric fluid

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ABSTRACT

Immersion cooling technologies have become an emerging cooling method for data center cooling. The studies available in open literature dealt with immersion cooling performance of data servers using mineral oil, EC-100, and Opticool 872552, but the studies focusing on the performance of FC-40 dielectric fluid is not available. Therefore, this study aimed to develop a single-phase liquid immersion cooling system using FC-40 dielectric fluid for a 1U (44.45 mm height) data server. The present study experimentally investigated the influence of different design and operating parameters, including the inlet/outlet port configuration (T and Z configurations), the bypass effect, the flowrate (1–3 LPM), the heating load (200–600 W), the inlet fluid temperature (15–35 °C), the presence of suction fan with various arrangements, and different heat sink bases (solid base, vapor chamber and heat pipe base). The results showed that the T-configuration could provide a 12.6% and 0.5–2.8 °C reduction in thermal resistance and case temperature compared to the *Z*-configuration, respectively. Furthermore, the heat sink with heat pipe and vapor chamber base offered a minimum case temperature of 56.7 °C and 55.2 °C against the 60 °C experienced in the heat sink with flat plate base, respectively.

1. Introduction

Data centers (DCs) are indispensable to modern computing infrastructure. Data racks, containing typical information technology (IT) devices, such as servers, storage, and network devices, are housed in DC to facilitate computing, processing, storage, and communication [1,2]. These devices consumes enormous electricity, which subsequently turns into heat that needs to be well managed for their reliable operation [3]. To tailor this problem, the IT devices must be operated with temperature being lower than their threshold. Hence, effective cooling of the data centers is a must-have measure [4,5]. Notice that currently the data centers consume almost 1–2% of total electricity, in which about 40% has been used on their cooling infrastructure [6,7]. In fact, the electricity expenditure for data centers has doubled in the past decade and are expected to be triple or even quadruple within the next decade [8,9]. In this regard, improving energy efficiency to reduce the carbon footprint becomes the primary concern of data centers in recent years. One of the major areas where energy consumption can be minimized is reducing cooling demands, which accounts for almost 40% of energy consumption for typical DCs.

For cooling technology applicable for DCs, the air-cooling method has been widely employed for its simplicity in deployment. Cooling demands up to 37 W cm^{-2} can efficiently be resolved through air-cooled systems [10]. However, beyond this limit, air-cooled systems may fall short of the cooling requirement. Note that in modern DCs the cooling demand can easily surpass this threshold due to high performance computing applications such as big data, cloud computing, artificial intelligence, smart and distributed manufacturing systems, autonomous vehicles, and smart energy systems [11]. Hence, the indirect liquid cooling methods like single-phase or two-phase cold plate were employed for high-power-density devices. The heat transfer coefficient of single-phase cooling is restricted to ~2000 W m^{-2} K^{-1}, while the two-phase method, depending on the working fluid, can offer much higher values [12]. Although indirect cooling provides a higher heat transfer rate, two-phase flow boiling methods suffer greatly from hydrodynamic instabilities and pressure drops [13] which normally require liquid pump. This also raises additional problems for higher energy consumption and reliability issues. Besides, the thermal interface materials in packing the liquid cold plate and electronic devices can also increase the thermal resistance appreciably. In essence, packaging constraints are

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the major challenges for both air-cooling and indirect liquid cooling methods [14]. However, immersion cooling can provide effective removal of the interface resistance, packaging constraints, and minimize the pumping requirements.

Immersion cooling technology has become a viable alternative for air-cooling and indirect cooling methods. In immersion cooling, the server is fully submerged in the dielectric fluid, facilitating direct contact between the IT devices (i.e., heat source) and the working fluid (i.e., cooling medium) [15]. This arrangement can eliminate the thermal contact resistance seen in indirect cooling, thereby improving the heat transfer rate. Moreover, packaging constraints are minimized in immersion cooling. In practice, immersion cooling can be deployed either in single-phase or two-phase mode. The latter adopts latent heat transfer (i.e., phase change phenomenon) and is especially suitable for ultra-high heat flux [16–18]. In two-phase cooling, fluorocarbon refrigerants are commonly used, which have a relatively higher global warming potential (GWP). Hydrofluoroethers (Novec 7000, Novec 7100, Novec 7200, and Novec 7300) have recently been developed to replace high GWP refrigerants. However, the HFE fluids still contains a GWP of 55–530 [19]. Furthermore, the condenser requirement makes two-phase immersion cooling systems more sophisticated. Therefore, the single-phase immersion cooling method has been developed to simplify the installation of immersion cooling. In single-phase immersion cooling, the working fluid is typically mineral oil, which has a relatively higher boiling point. The studies related to single-phase immersion cooling using dielectric oil are summarized in Table 1.

Eiland et al. [20] developed a single-phase immersion cooling facility for an Intel-based open server with two central processing units (CPU). The power density of each CPU is about 95 W. The flowrate (V_{fluid}) and the inlet fluid temperature ($T_{fluid,in}$) control the server temperature. The power consumption can be minimized at a high $T_{fluid,in}$ and a low flowrate, and the partial power usage effectiveness can be minimized to 1.03 using mineral oil. Bansode et al. [21] studied the operating behavior of oil immersion cooling at different ambient air temperatures (T_{air} = 25 °C to 55 °C and RH = 50%). The power consumption of the server remained identical under all ambient temperatures, while the junction temperature increased with the air temperature, and it reached the maximum value at T_{air} = 55 °C. Similarly, Bansode et al. [22] examined the performance of oil-submerged servers under extreme ambient conditions (T_{air} = −20 °C to 10 °C and RH = 100%). The server performed well under extreme ambient temperatures and relative humidities with an average power variation of 20 W. In extreme weather conditions, no oil leakage or other starting problems are found.

Gandhi et al. [23] compared the performance of mineral oil and EC-100 with the air-cooling method. For a fixed inlet temperature and flowrate, the use of mineral oil offered a lower junction temperature, followed by EC-100 and air. Moreover, the operating conditions, such as the oil flowrate and inlet temperature, to maintain the junction temperature below its threshold value are entirely different amid test oils. Shinde et al. [24] found a maximum of 16.83% reduction in server power with oil when compared to air-cooling. Shah et al. [25] compared the performance of different heat sink configurations using numerical analysis under single-phase oil immersion cooling conditions. The heat transfer behavior of plate-fin heat sinks and pin-fin heat sinks is identical to that of conventional parallel plate heat sinks. Shah et al. [26] compared the performance of oil-cooled servers with different open rack units such as 1U, 1.5U, and 2U. The 1.5U open rack unit server outperformed 1U and 2U servers at a 40 °C inlet oil temperature and 1 LPM oil flowrate.

The above studies mainly focused on single-phase immersion cooling using dielectric oil. Recently, Cheng et al. [27] used the Novec 7100 dielectric fluid for single immersion cooling. They concluded that the higher flowrate provides a more uniform temperature distribution around the heat sink. However, the temperature distribution on the right and left sides of the heat sink is quite different (asymmetry). This implies

Table 1
Summary of single-phase immersion cooling studies.

Reference	Fluid	Server	Heat capacity	Process parameters	Findings
Eiland et al. [20]	Mineral Oil	Intel server with 2 CPU	95 W/CPU (TDP)	V_{fluid} = 0.5–2.5 LPM	1. A partial power usage effectiveness of 1.03–1.17 can be obtained
				$T_{fluid,in}$ = 30–50 °C	2. Server can safely be operated up to $T_{fluid,in}$ = 45 °C
Bansode et al. [21]	Opticool 872552	Intel Core i7 5th Gen with Quad core processor	100–110 W	$T_{fluid,in}$ = 30–60 °C	1. Oil immersion cooling can safely be operated between T_{air} = 25 °C and 55 °C
				T_{air} = 25–55 °C	2. Maximum junction temperature is obtained at T_{air} = 55 °C
Bansode et al. [22]	Opticool 872552	Intel Core i7 5th Gen with Quad core processor	101–110 W	$T_{fluid,in}$ = 30–60 °C	1. Server can operate well under extreme ambient conditions with 20 W power variations
				T_{air} = −20 to +55 °C	2. No leakage and start troubles observed
Gandhi et al. [23]	EC-100	HP ProLiant DL160 G6 1U	500 W	V_{fluid} = 0.5–3 GPM $T_{fluid,in}$ = 15–25 °C	1. For fixed inlet temperature and flowrate, mineral oil showed a low junction temperature, followed by EC-100 and air
	Mineral Oil			V_{fluid} = 0.5–1 GPM $T_{fluid,in}$ = 25–45 °C	
Shinde et al. [24]	Mineral Oil	HP ProLiant DL160 G6	95 W/CPU (TDP)	V_{fluid} = 0.5–1 GPM	1. A partial power usage effectiveness of 1.093–1.1 can be obtained
				$T_{fluid,in}$ = 25–45 °C	2. Server can safely be operated up to $T_{fluid,in}$ = 45 °C
Shah et al. [25]	Mineral Oil and EC-100	Intel 3rd Gen server	95 W/CPU (TDP)	V_{fluid} = 0.5–2.5 LPM $T_{fluid,in}$ = 30–50 °C	1. Performance of plate fin, pin fin, and parallel plate heat sinks is similar under oil immersion cooling
Shah et al. [26]	Mineral Oil and EC-100	Intel 3rd Gen server	115 W/CPU	V_{fluid} = 0.5–2.5 LPM	1. The 1.5U open rack unit server performed better than 1U and 2U
				$T_{fluid,in}$ = 30–50 °C	2. EC-100 provided a

(continued on next page)

Table 1 (*continued*)

Reference	Fluid	Server	Heat capacity	Process parameters	Findings
					lower server temperature over mineral oil
Cheng et al. [27]	Novec 7100	Intel server	95 W/CPU (TDP)	Oil velocity = 0.4–1.2 m/s	1. Aluminum and copper heat sinks are compatible with the working fluid 2. A higher flowrate caused a more uniform temperature distribution

that the existing server packaging may not be suitable for immersion cooling due to the different flow distribution across the heat sink induced by dielectric oil. As a result, some new development in server architecture and heat sink design suitable for immersion cooling are needed. Furthermore, they found that the aluminum and copper heat sinks are compatible with the Novec 7100 dielectric fluid.

Shah et al. [28] discussed the various non-thermal aspects to be considered while using oil immersion cooling such as flammability, chemical stability, material compatibility, dielectric properties, and safety. The accelerated thermal cycling test of mineral oil showed that the ramp time and dwell period of the oil are significantly different from those of air. The results implicate the ASHRAE thermal guidelines pertaining to air-cooled data centers are inappropriate to oil immersion cooling, enforcing new standards for immersion cooling. In another study, Shah et al. [29,30] compared the reliability aspects of oil immersion cooling with the air-cooling method. They concluded that oil immersion cooling showed better reliability over air cooling for the following reasons,

- Reduced working temperature of IT components
- Uniform temperature distribution
- Oxidation and corrosion caused by air is eliminated
- Noise-free operation due to the absence of moving parts, such as fans
- The accumulation of dust and contaminants is avoided
- Insensitive to humidity and temperature of air
- Elimination of ducting
- Reduced corrosion and electrochemical migration
- Elimination of zinc and tin whisker formation

In summary, single-phase immersion cooling using dielectric fluid offers various advantages over conventional air-cooling methods in terms of heat transfer and energy-saving aspects. However, based on the above literature review, studies related to single-phase immersion cooling using dielectric fluid are limited. Additionally, the previous studies mainly examined the performance of mineral oil, EC-100, and Opticool 872552. However, studies concerned with other dielectric fluids such as EC-110, FC-40, and PAO 8 are limited in the open literature. Furthermore, the prior studies were mainly focused on optimizing process parameters such as fluid inlet temperature and flowrate. However, the influence of heat sink design parameters (e.g., fin pitch), inlet/outlet arrangements, and bypass effect has not been studied in the past. Therefore, the present study aimed to investigate the performance of single-phase liquid immersion cooling system of 1U server using FC 40 under different operating conditions, including heat load, flowrate, and fluid inlet temperature. Furthermore, this study also analyzes the influence of fin pitch, bypass effect, and inlet/outlet configuration on cooling system performance.

2. Experimental setup and numerical methodology

2.1. Experimental setup

The schematic of the single-phase liquid immersion cooling experimental facility is shown in Fig. 1. The main components of the test facility are: (1) data server (1U), (2) tank, (3) pump, (4) flow meter, and (5) thermostatic bath. The data server comprises a thermal test vehicle (TTV) from Intel and a 3D printed ram mounted on the printed circuit board (PCB). Note that the 3D printed RAMs are attached to the PCB to simulate the typical flow behavior, but the heat load from the RAMs is not considered in this study as the corresponding heat generation is minimum compared to the central processing unit (CPU). The dimensional details of the 3D printed RAM can be found in Fig. 1(d). The TTV is known as a simulated CPU, which simulates the heat load equivalent to that of the CPU and simultaneously measures its temperature. The TTV used in this study were compartmentalized into 4 dies where the heating power can be modulated, as shown in Fig. 1(e). The resistors were embedded to the copper plane inside the TTV to provide the necessary heating power to the TTV, and the heat can be distributed across the TTV case due to its thermal mass. The TTV was powered using an external DC power supply that is connected to the PCB using connectors. The heat sink (made up of copper) was firmly attached to the TTV using 4 screws, as shown in Fig. 2(a). The heat sink comprised of base and fins. The thickness of the heat sink base was 4.5 mm, and the thickness and height of the fins were taken as 0.25 mm and 20.9 mm, respectively. To minimize contact resistance, thermal grease (TC-5888; $k = 5.2\ \mathrm{W\ m^{-1}\ K^{-1}}$) was applied between the heat sink and TTV base. An Intel-based server (1U) with a heat sink was fully submerged in dielectric fluid (FC-40) filled in the sealed tank, as shown in Fig. 1(b). The dimensional details of the data server, heat sink, and tank can be seen in Fig. 2. The dielectric fluid was pumped into the tank that contains the 1U server, and removes the heat from the server. Subsequently, the heated dielectric fluid flowed back to a thermostat bath where it was cooled to a prescribed temperature. Upon cooling the dielectric fluid, it was circulated back to the server through a filter which helps to filter particulates and dust present in the fluid. The flowrate was controlled by flow regulators, and it was measured through a flow meter (BNC Industrial: BH20-0F5-BSP). It should be noted that the entire experimental system was insulated to minimize heat loss. The details of the measuring instruments are listed in Table 2.

The performance of the server is satisfactory if its case/chip temperature (T_{case}) is below a threshold value (<80 °C). Therefore, two T-type thermocouples were embedded on the TTV surfaces to monitor its temperature performance. Additionally, the temperature of the dielectric fluid was measured at the inlet ($T_{fluid,in}$) and outlet ($T_{fluid,out}$) of the tank using T-type thermocouples, as shown in Fig. 1(a). Furthermore, the input power ($Q_{in} = VI$) to the server was calculated through voltage (V) and current (I) measured using the DC power supply (6230K-150). Since the system is fully insulated to minimize heat loss, the heat loss between the supply power (Q_{in}) and heat transfer rate (Q) was negligible. The overall thermal resistance (R) of the cooling system can be estimated using the following relationship,

$$R = \frac{T_{case} - T_{fluid,in}}{Q_{in}} \tag{1}$$

The experimental uncertainty consists of test and systematic errors, and it can be estimated by [31,32]:

$$R_{\alpha} = \sqrt{\left[\left(\frac{\partial \alpha}{\partial y_1} Z_1\right)^2 + \left(\frac{\partial \alpha}{\partial y_2} Z_2\right)^2 + \ldots + \left(\frac{\partial \alpha}{\partial y_n} Z_n\right)^2\right]} \tag{2}$$

where the parameter (α) is a function of independent variables ($y_1, y_2, \ldots, y_n$) and their corresponding uncertainties of independent variables are $Z_1, Z_2, \ldots, Z_n$. The maximum uncertainty of the thermal resistance was

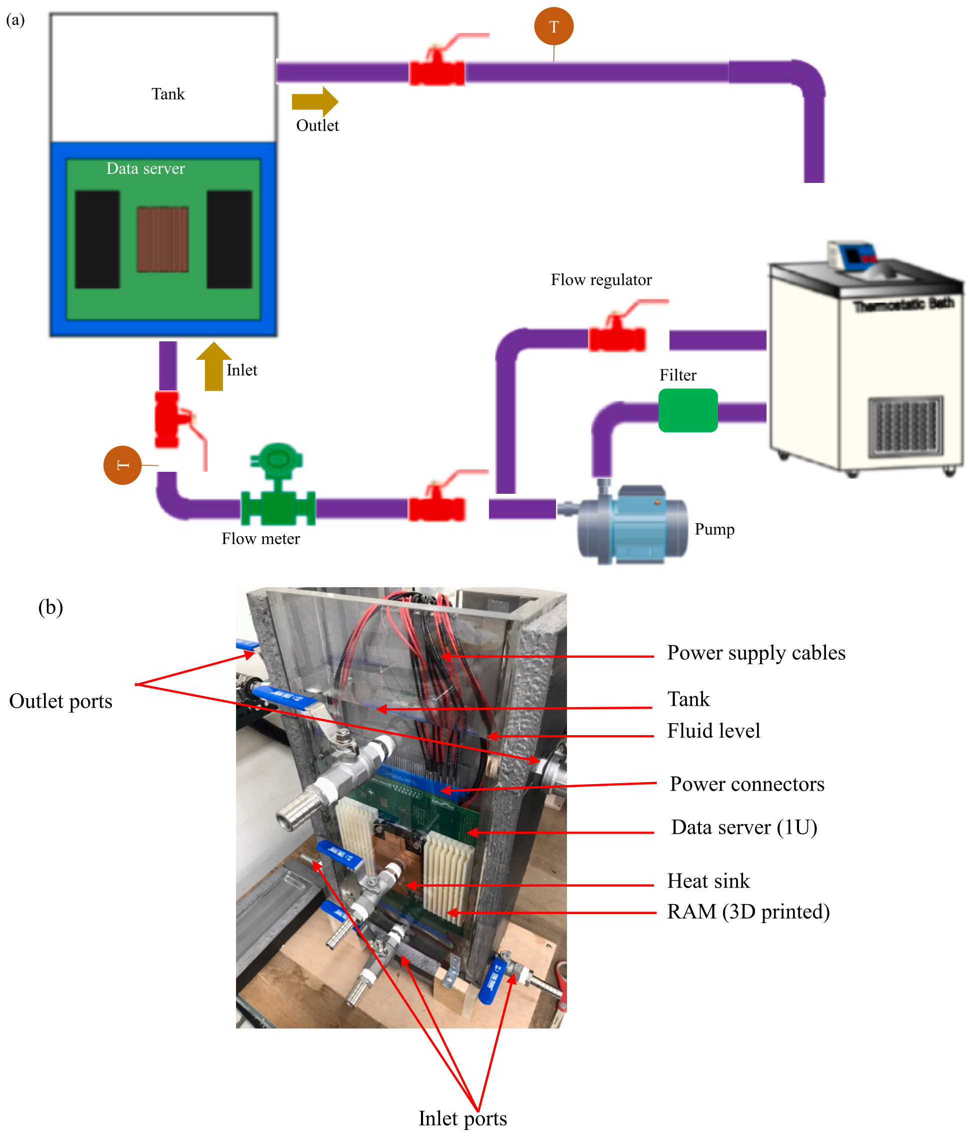

Fig. 1. (a) Schematic of the experimental setup and (b) Photo showing the data server immersed in FC-40.

estimated to be 6.8%.

The performance of the single-phase immersion cooling system can be primarily affected by various design and operating factors, including the fin pitch, flowrate, fluid inlet temperature, and the inlet/outlet configurations. Therefore, the experiments were carried out for various fin pitch, flowrate, fluid inlet temperature, and inlet/outlet port configurations to identify the optimal parameters that can provide efficient cooling, as listed in Table 3. Furthermore, the important thermophysical properties of FC-40 (3M Fluorinert Electronic Fluid FC-40, USA) are listed in Table 4.

2.2. Numerical methodology

The results presented in terms of thermal resistance and case temperature in section 3 are entirely based on the experimental works; however, the numerical simulation was carried out to understand the flow behavior of the working fluid within data server. At the inlet and outlet, the flowrate and pressure boundary conditions were enabled, and the heat input was given at the base of the heat sink (Refer Fig. 2 (c)). The no-slip boundary conditions prevailed at the solid surfaces. The Ansys Fluent software was used for simulation, and the laminar flow conditions prevailed. The steady state, three dimensional, and incompressible flow assumptions were considered in this study. Besides, the fluid properties were given as a function of temperature in the Ansys software, and the gravity effects were also included in the simulation. The continuity, momentum, and energy equations are simultaneously solved to predict the flow flied and temperature distribution in the data server cooling module. The continuity, momentum, and energy equations are listed below,

Continuity equation,

$$\nabla\cdot\left(\rho\vec{V}\right) = 0 \tag{3}$$

Momentum equation,

$$\rho\left(\vec{V}\cdot\nabla\vec{V}\right) = -\nabla P + \nabla\cdot\left(\mu\left(\nabla\vec{V}\right)\right) + \rho\vec{g} \tag{4}$$

Energy equation,

$$\nabla.\left(\vec{V}T\right) = \alpha\nabla^2 T \tag{5}$$

where ρdenotes the density of the working fluid (kg m^{-3}), P represents the pressure (Pa), $\mu$is the dynamic viscosity (Pa s), Tindicates the temperature (K), αdenotes the thermal diffusivity (m^2 s^{-1}), $\vec{g}$indicates the acceleration due to gravity (m s^{-2}), and $\vec{V}$ is the velocity vector (m/ s).

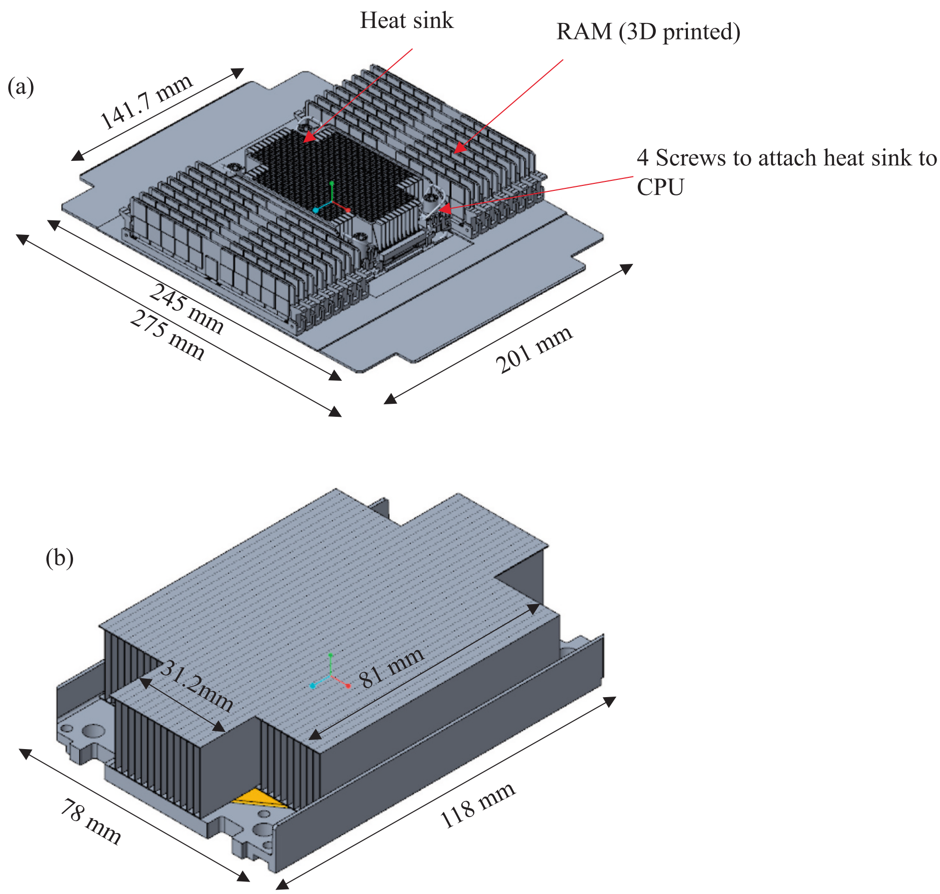

Fig. 2. Dimensional details of the (a) data server, (b) heat sink, (c) 1U-tank, (d) RAM, and (e) thermal test vehicle.

The grid independent study was performed to confirm the mesh sensitivity on the results, and the results are presented in Table 4. The maximum case temperature was considered as a parameter to verify the mesh sensitivity. According to Table 5, the case temperature showed an insignificant variation when the number of mesh elements was increased above 7.7 million, which was adopted for further numerical simulation.

3. Results and discussion

As mentioned earlier, the performance of the single-phase liquid immersion cooling system depends on various parameters, including the inlet/outlet configuration, bypass effect, suction fan, flowrate, heat load, and inlet temperature of the fluid. The influence of those parameters is sequentially discussed in the following sections subject to different fin pitches (1.2 and 2.2 mm, respectively). Vapor chamber and heat pipe are integrated at the base of heat sink for effective heat spreading, which is also examined in this study.

3.1. Inlet/outlet configuration

Depending on the inlet and outlet port arrangements, multiple inlet/outlet configurations are applicable for single-phase immersion cooling. In this study, two inlet/outlet arrangements are studied, including (1) *Z*-configuration and (2) T-configuration, as shown in Fig. 3. In the Z-configuration, the dielectric fluid enters the at the bottom of left-hand side (blue arrow) and leaves from the top of the right-hand side (red arrow). Whereas, in the T-configuration, the fluid enters from the base of the tank and leaves from the two sideways (left and right) at the top of the tank.

The overall thermal resistance and case temperature for different inlet/outlet configurations are presented in Fig. 4. In general, the T-configuration outperformed the *Z*-configuration for a prescribed flowrate. A maximum of 12.6% reduction in overall thermal resistance is obtained by the T-configuration over the Z-one. Furthermore, the case temperature of the T-configuration is about 0.5–2.8 °C lower than that of the Z-arrangement. It is interesting to note that the thermal resistance and case temperature decrease with flowrate for the T-configuration, whereas no appreciable change of thermal resistance with the rise of flowrate for Z-configuration. Specifically, in the Z-arrangement, the case temperature at 1 and 3 LPM is about 58.8 and 58.6 °C, respectively. The main reason for this trend is the fluid flow path, as shown in an accompanied computational fluid dynamics (CFD) simulation in Fig. 5. In the case of the Z-arrangement, most of the fluid enters from the side inlet port eventually bypass the heat sink without actively participating in heat removal, as shown in Fig. 5(a). As a result, the cooling performance of the server (R and T_{case}) is independent of flowrate. Opposed to the Z-arrangement, fluid enters the tank from the inlet port at the base (located straight below the heat sink) revealing a jet-impingement flow characteristic upon the heat sink. The impingent phenomenon becomes more pronounced with the flowrate. Hence, the heat transfer performance of the T-arrangement is superior to Z-arrangement, yielding a lower thermal resistance and a lower case temperature accordingly.

3.2. Bypass effect

While designing the 1U tank for the single-phase cooling system, it is necessary to consider the gap between the heat sink and the wall of the 1U tank wall (inner wall), as shown in Fig. 6. To analyze the influence of the gap between the heat sink and the 1U tank on cooling performance, this study examines three different gaps of 5 mm, 2 mm, and 0 mm (i.e.,

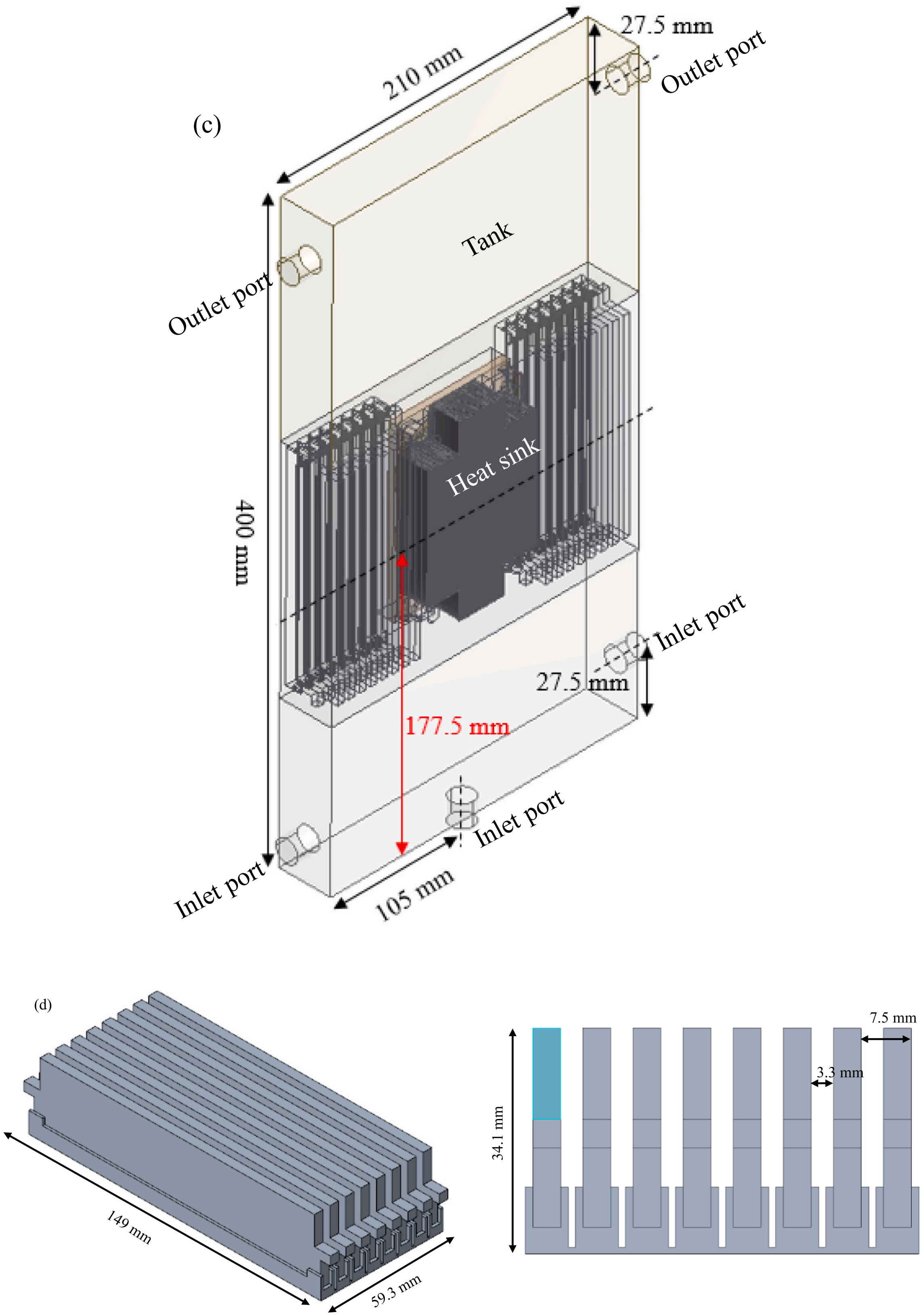

Fig. 2. (*continued*).

no-gap). The results of these three different scenarios are presented in Fig. 7. For both fin pitches (1.2 and 2.2 mm), the presence of gap degrades the cooling performance. The thermal resistance is increased by approximately 3% and 7.4% when the gap is increased from 0 mm to 5 mm for the 2.2 mm and 1.2 mm fin pitches, respectively. Similarly, the case temperature is increased about 0.7–1.5 °C when the gap is

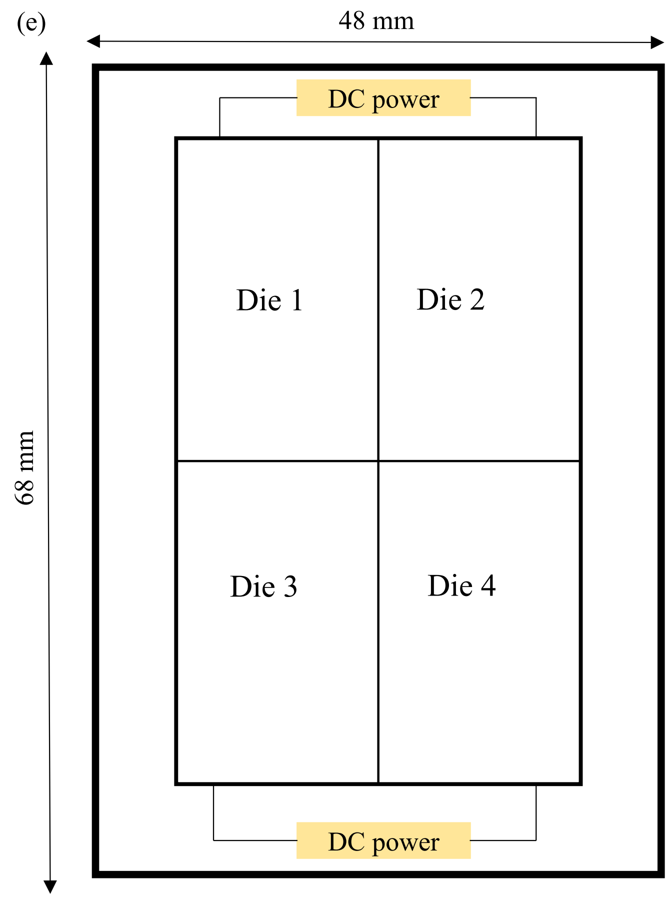

Fig. 2. (*continued*).

Table 2
Details of the measuring instruments.

Parameter	Instrument	Range	Accuracy
Fluid temperature	T-type thermocouple	−200–350 °C	±0.1 °C
Case temperature	T-type thermocouple	−200–350 °C	±0.1 °C
Fluid flowrate	Flow meter	20–1300 LPH	±0.50%
Voltage	DC power supply - Voltmeter	0–150 V	±0.05% F.S.
Current	DC power supply - Ammeter	0–20 A	±0.1% F.S.

Table 3
Test conditions.

Description	Test range
Heat sink's fin pitch	1.2 mm and 2.2 mm
Flowrate	1–3 LPM
Fluid inlet temperature	15–35 °C
Heat load	200–600 W
Inlet/outlet configuration	T and Z arrangements

increased to 5 mm from 0 mm. The performance drop is associated with larger bypass flow upon a larger gap as depicted from the numerical simulation in Fig. 8. From Fig. 8(a), with the presence of gap between the heat sink and 1U tank, the fluid tends to flow through the gap because of the lower fluid flow resistance. This bypass fluid stream does not actively participate in heat transfer process. Consequently, a higher thermal resistance prevails and results in a larger case temperature, as shown in Fig. 8(c). If the gap is fully eliminated, the fluid can pass through the heat sink more effectively as shown in Fig. 8(b), therefore yielding a lower thermal resistance and case temperature as illustrated in Fig. 8(d). In this regard, the no-gap conditions were adopted for

Table 4
Thermophysical properties of FC-40.

Properties	Values
Density (@ 16 °C)	1874.44 kg m^{-3}
Latent heat	69 kJ kg^{-1}
Boiling point (@ 1 bar)	165 °C
Coefficient of thermal expansion	0.0012 K^{-1}
Specific heat	
0 °C	1014 J kg^{-1} K^{-1}
40 °C	1076 J kg^{-1} K^{-1}
100 °C	1169.4 J kg^{-1} K^{-1}
Kinematic viscosity	
0 °C	$5e^{-6}$ m^2 s^{-1}
40 °C	$1.5e^{-6}$ m^2 s^{-1}
100 °C	$5.4e^{-7}$ m^2 s^{-1}
Thermal conductivity	
0 °C	0.067 W m^{-1} K^{-1}
40 °C	0.0642 W m^{-1} K^{-1}
100 °C	0.0601 W m^{-1} K^{-1}

Table 5
Grid independent study results.

Mesh elements	Max case temperature (K)	Error (%)
935,980	367.60	12.07
1,986,415	359.70	9.66
7,729,868	327.80	0.06
13,005,656	328.00	Reference

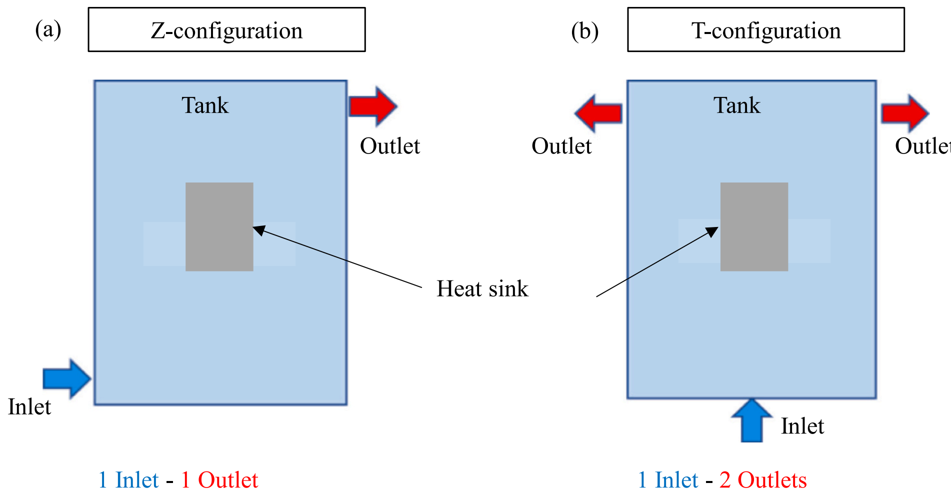

Fig. 3. Inlet/outlet port arrangements, (a) *Z*-configuration with 1 inlet (at the bottom of the left side) and 1 outlet (at the top of the right side) and (b) T-configuration with 1 inlet (at the base) and 2 outlets (at the top of the left and right sides).

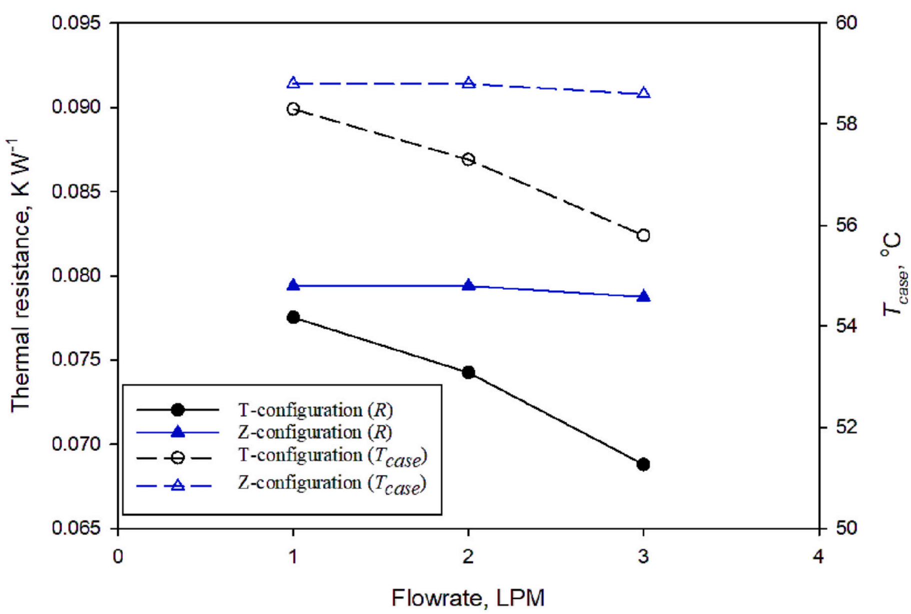

Fig. 4. Effect of inlet/outlet configuration on the cooling performance (Fin pitch = 1.2 mm, $T_{fluid,in}$ = 35°C, and Q_{in} = 300 W).

further experimental analysis in the subsequent sections.

3.3. Influence of operating conditions on cooling performance

The main operating conditions affecting the performance of single-phase immersion cooling systems are: (1) flowrate, (2) heat load, and (3) fluid inlet temperature. This section addresses the effect of these parameters on the thermal resistance and case temperature, as depicted in Figs. 9-11.

The effect of flowrate on the thermal resistance and case temperature is presented in Fig. 9. As expected, both thermal resistance and case temperature decreased with flowrate as a result of higher flow inertia. For 2.2 mm pitch, the resistances and the case temperature are decreased by 15% and 3.9 °C when the flowrate increases from 1 LPM to 3 LPM, respectively. Similarly, for 1.2 mm pitch, the resistance and the case temperature are reduced by 11% and 2.5 °C, respectively. Note that under all the flowrates, the performance of the 1.2 mm fin pitch is better than that of 2.2 mm case due to the increased surface area in heat sink with 1.2 mm fin pitch. The heat sink with 1.2 mm fin pitch yielded a 12–16% and 2.9–4.3 °C reduction in the thermal resistance and case temperature compared to 2.2 mm, respectively.

The influence of heating load on the performance of single-phase immersion cooling is depicted in Fig. 10. The thermal resistance is decreased with an increase in heating load for both fin pitches. Whereas the case temperature is increased with heating load. At a flowrate of 3 LPM, the single-phase immersion cooling system using FC-40 can maintain the case temperature below 80 °C up to a heat load of 600 W. Under all heat loads (up to 600 W), the 1.2 mm fin pitch provided a 9–15% lower thermal resistance over the 2.2 mm case. Similarly, the heat sink with 1.2 mm fin pitch experienced a 1.6–6.8 °C lower case temperature than its counterpart. Note that the heating load imposes a negligible influence on the thermal resistance of a typical air-cooled heat sink due to minor variation in the viscosity of working fluid (i.e., air) with temperature. Whereas an appreciable reduction in the thermal resistance with rising heat load is encountered for the present FC-40 liquid cooled heat sinks. This is because of the considerable reduction in viscosity for FC-40 with the rise of heating load (or temperature), thereby increasing the effective Reynolds number across the heat sink

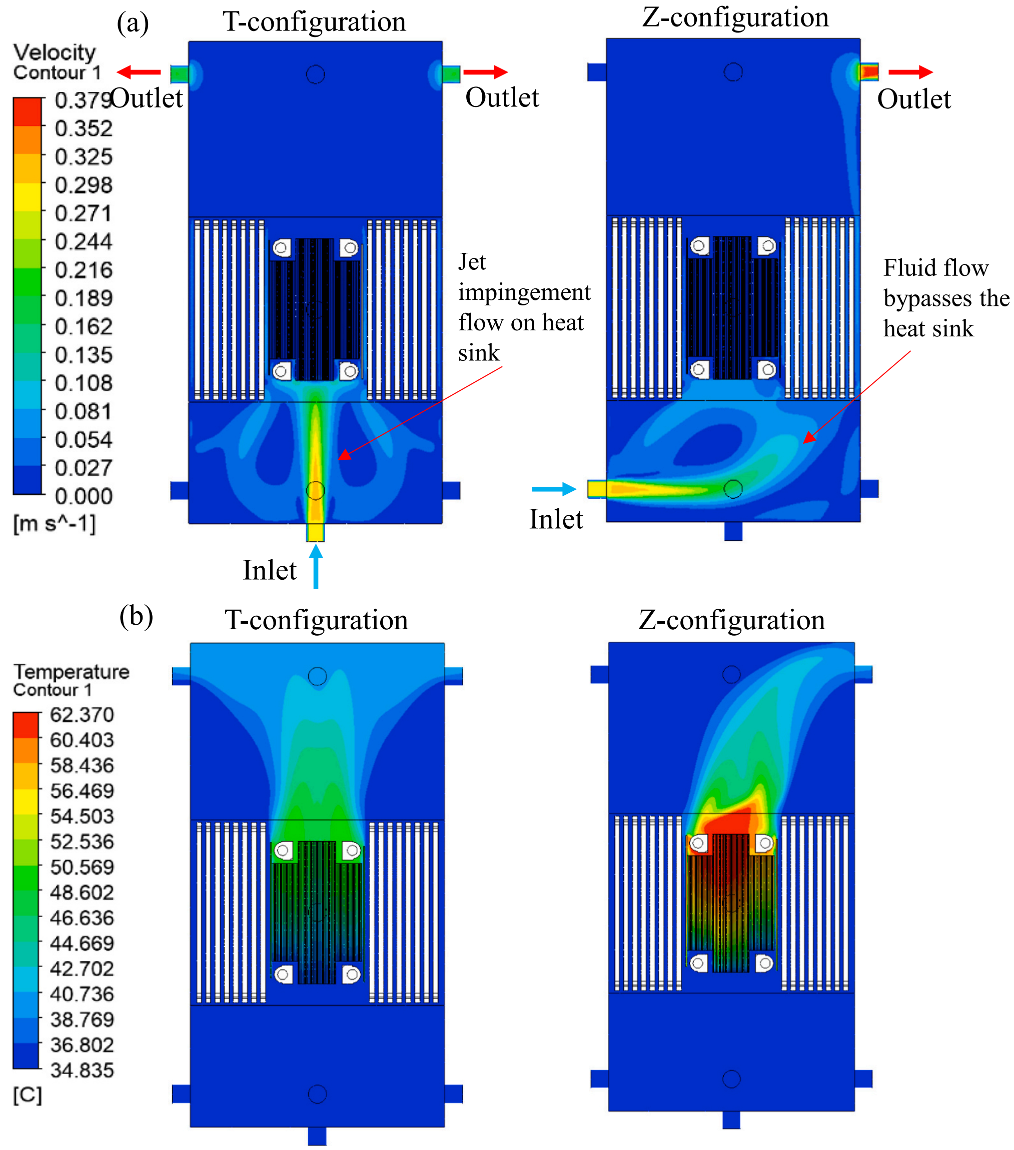

Fig. 5. The fluid flow behavior in the T and Z configurations (a) Velocity distribution and (b) Temperature distribution.

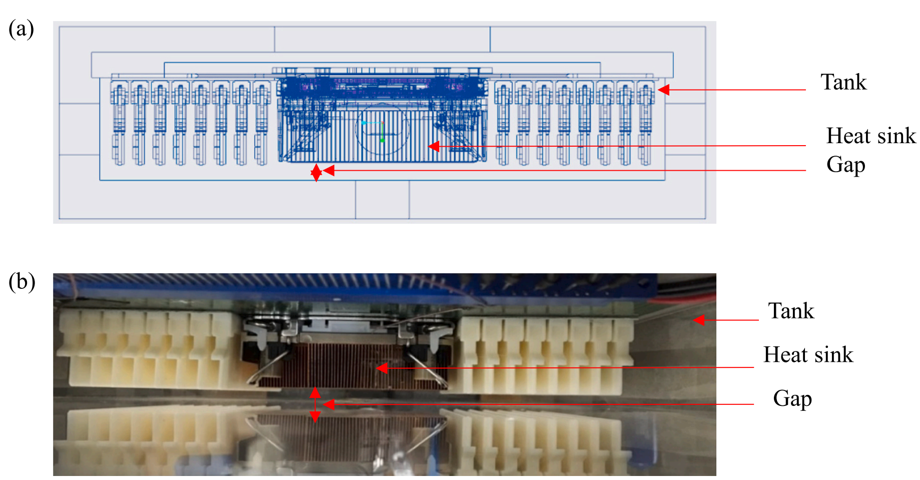

Fig. 6. The gap between the heat sink and the tank (a) schematic representation and (b) pictorial representation.

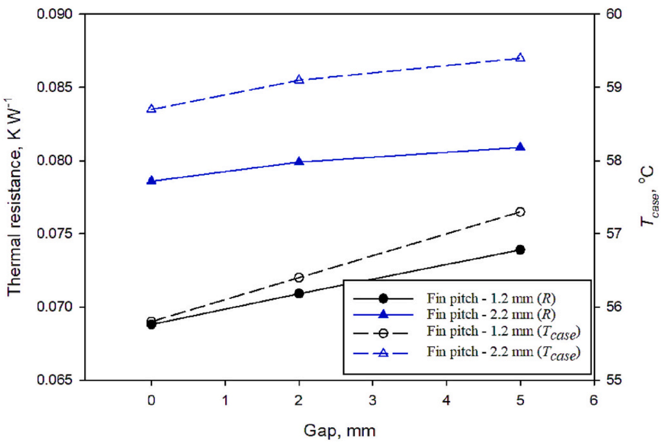

Fig. 7. Influence of the gap between the heat sink and the 1 U tank on cooling performance. (T-configuration, Flowrate = 3 LPM, $T_{fluid,in}$ = 35°C, and Q_{in} = 300 W).

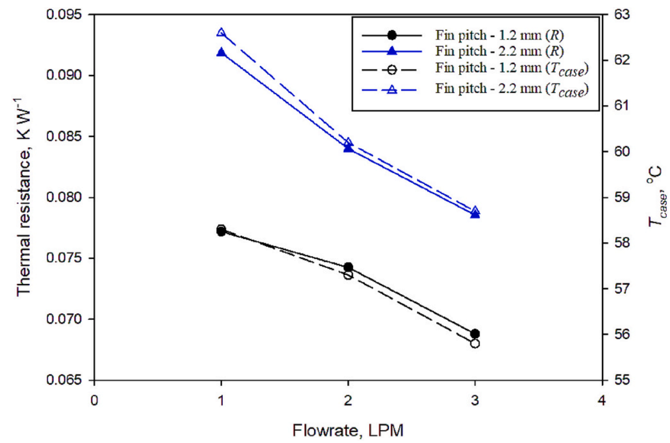

Fig. 9. Effect of flowrate on cooling performance. (T-configuration, $T_{fluid,in}$ = 35°C and Q_{in} = 300 W).

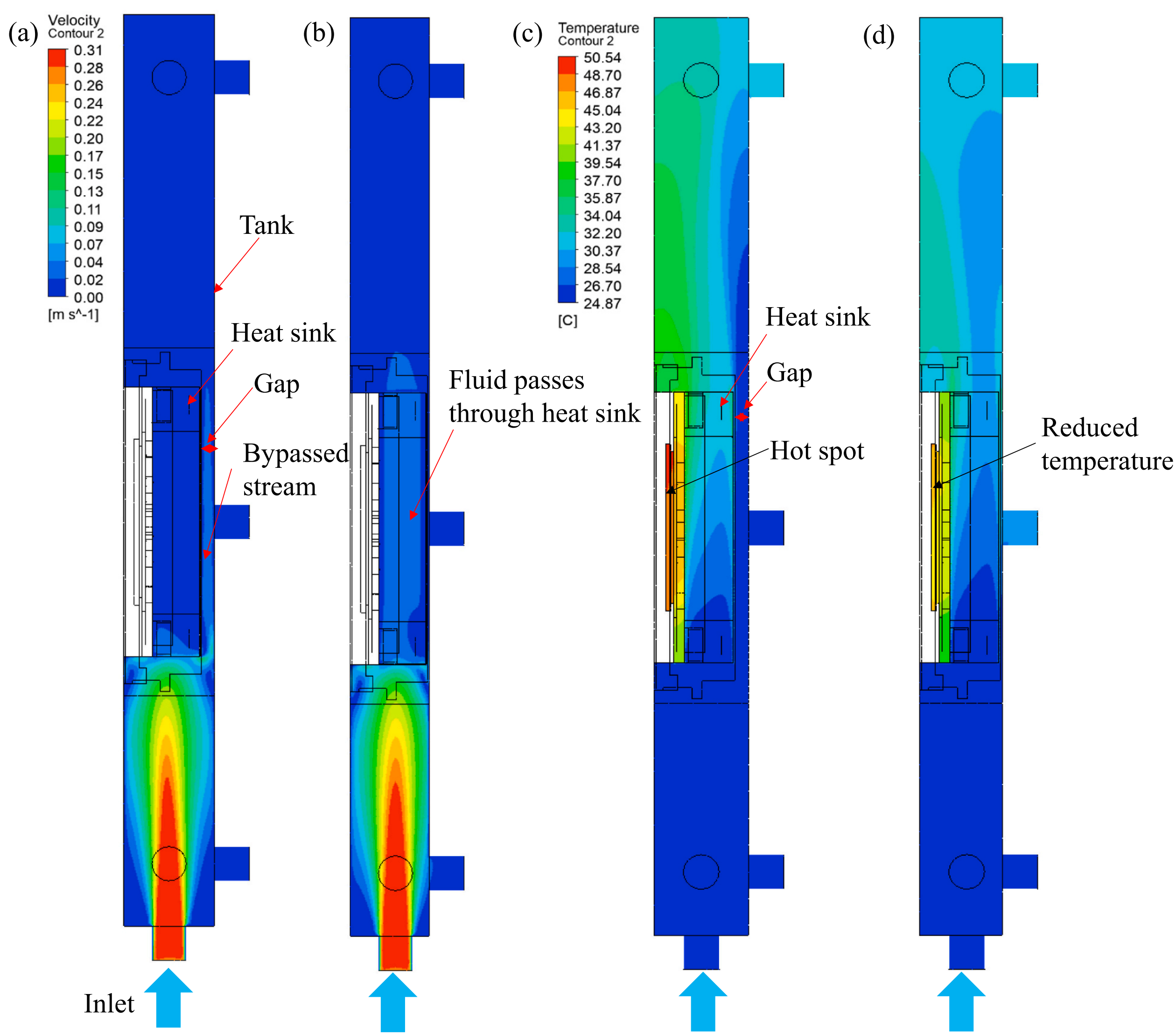

Fig. 8. Bypass effect on velocity and temperature distribution, (a) Velocity distribution - Gap = 5 mm (b) Velocity distribution – Gap = 0 mm, (c) Temperature distribution – Gap = 5 mm, and (d) Temperature distribution – Gap = 0 mm. (T-configuration, Flowrate = 3 LPM, Q_{in} = 300 Wand $T_{fluid,in}$ = 35°C).

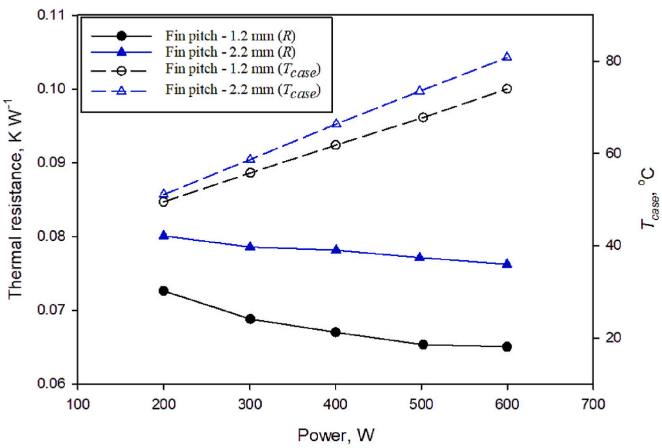

Fig. 10. Performance of single-phase immersion cooling system under different heat loads. (T-configuration, Flowrate = 3 LPM, and $T_{fluid,in}$ = 35°C).

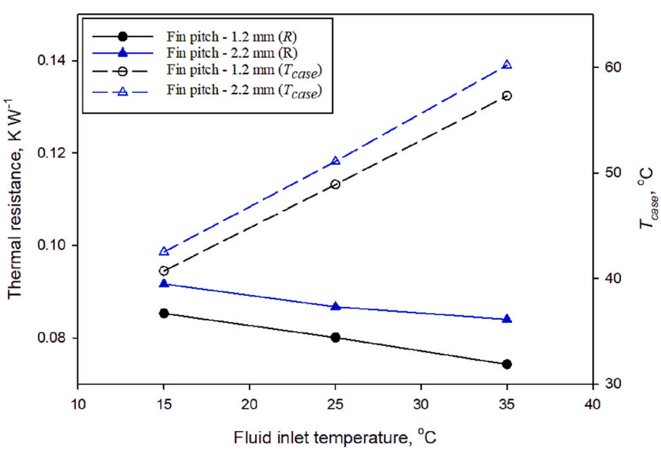

Fig. 11. Influence of fluid inlet temperature on thermal resistance and case temperature. (T-configuration, Flowrate = 2 LPM, and Q_{in} = 300 W).

and promoting the heat transfer performance accordingly.

The performance of a single-phase immersion cooling system subject to inlet temperatures is presented in Fig. 11. It is found that the thermal resistance is decreased with the increase of inlet temperature. Apparently, this is also attributed to the reduction in fluid viscosity. This can be made clear from Table 4 that the viscosity of the fluid is greatly reduced with the rise of temperature. Similarly, the corresponding effective Reynolds number is increased, which aids in augmenting heat transfer and yielding a low thermal resistance. Despite the thermal resistance decreases with the rise of temperature, the case temperature is increased with increasing inlet temperature. This is because of the reduction in effective temperature difference ($T_{case} - T_{fluid,in}$) between the fluid and heat source, thereby yielding a lower heat transfer rate and a higher case temperature ($Q = hA(T_{case} - T_{fluid})$).

3.4. Effect of installing suction fan downstream of the heat sink

In the previous sections, the dielectric fluid was circulated in the tank with the help of pump only. This type of conventional arrangement could cause an ineffective fluid flow across the heat sink and server (i.e., CPU, RAM, transistors, and among others). To entrain more fluid across the heat sink, suction fans are mounted at the top of the heat sinks (i.e., at the exit of heat sink). Note that the rated speed of the suction fan is nearly 13,000 rpm in air. However, the fan speed can be lower than its rated value when it is operated under FC-40 liquid condition due the higher viscosity of liquid. Though this manipulation, more fluid is directed to pass through the heat sink. Four different fan arrangements are studied for further elaboration of suction fans on the performance of single-phase liquid immersion cooling systems. The four different fan arrangements are listed in Table 6 and schematically presented in Fig. 12.

Table 6
Different fan arrangements.

Case No	Description
Case 1	No fan (see Fig. 12(a))
Case 2	Single fan (see Fig. 12 (b))
Case 3	Two fans with series arrangement (see Fig. 12(c))
Case 4	Two fans with parallel arrangement (see Fig. 12(d))

In Section 3.1 (i.e. without fan condition), it was seen that the T-configuration outperformed the *Z*-configuration due to the jet-impingement flow pattern caused by the T-configuration. Yet, the Z-configuration suffered considerably with the bypass flow. The presence of suction fan can minimize the influence of bypass. Hence, both configurations are taken for further comparison of the fan installation. The thermal resistance of different fan arrangements for both T and Z configurations is compared in Fig. 13. Similar to those without fan installation (i.e., the conventional one), the performance of T-configuration is better than those of Z-arrangement for all fan arrangements. Comparatively, the Z-configuration performance is improved further upon fans installation. For example, for test without fans, the thermal resistance and case temperature of the Z-configuration are about 13% and 2.8 °C higher than those of the T-configuration, respectively. Whereas, for case 4 (2 fans with parallel arrangement), the Z-configuration experienced just a 7% and 1.4 °C rise in resistance and temperature over the T-configuration, respectively, which is 2 times better than those without fan installation.

Among the four different fan arrangements, two fans in parallel connection with heat sink offered a minimum thermal resistance and a least case temperature. Compared to the no-fan condition, the case 4 (2 fans with parallel arrangement) offered a maximum of 11% reduction in thermal resistance for the T-configuration. Similarly, the case temperature is reduced by 2.5 °C from 55.8 °C (under no-fan condition) to 53.3 °C when two fans are installed in parallel at the exit of the heat sink for the T-configuration.

3.5. Vapor chamber and heat pipe base for heat sinks

In the previous sections, the experimental results were obtained for the conventional stacked fin heat sink design that contains a solid base plate (i.e., a flat plate). Usually, the thermal spreading resistance is higher for the heat sinks with a solid plate because of considerable spreading resistance from the smaller heat source (e.g. CPU). To minimize the spreading resistance, Ong et al. [33] proposed to increase the thickness of the base plate and the height of the fins. These solutions are not feasible for the heat sinks used in space-constrained data servers such as 1U server. Therefore, alternative solutions of the vapor chamber base and the embedded heat pipe base are proposed in this study to minimize the spreading resistance.

In the case of the vapor chamber case, the conventional base (i.e., solid plate) of the heat sink is replaced with a vapor chamber. The details of the vapor chamber base are shown in Fig. 14(a). The vapor chamber consists of evaporator, condenser, and vapor core sections. The total thickness of the vapor chamber is 3 mm. The thickness of the evaporator and condenser wall is approximately 1 mm and 0.5 mm, respectively. And the thickness of the evaporator and condenser wick is about 0.3 mm and 0.2 mm, respectively. The wick is made up of sintered copper powder with an average size of 150 μm and a porosity of 50%. Note that the area (8670 mm^2) of the vapor chamber base is identical to that of the solid flat plate base. The details of the heat pipes embedded in the heat

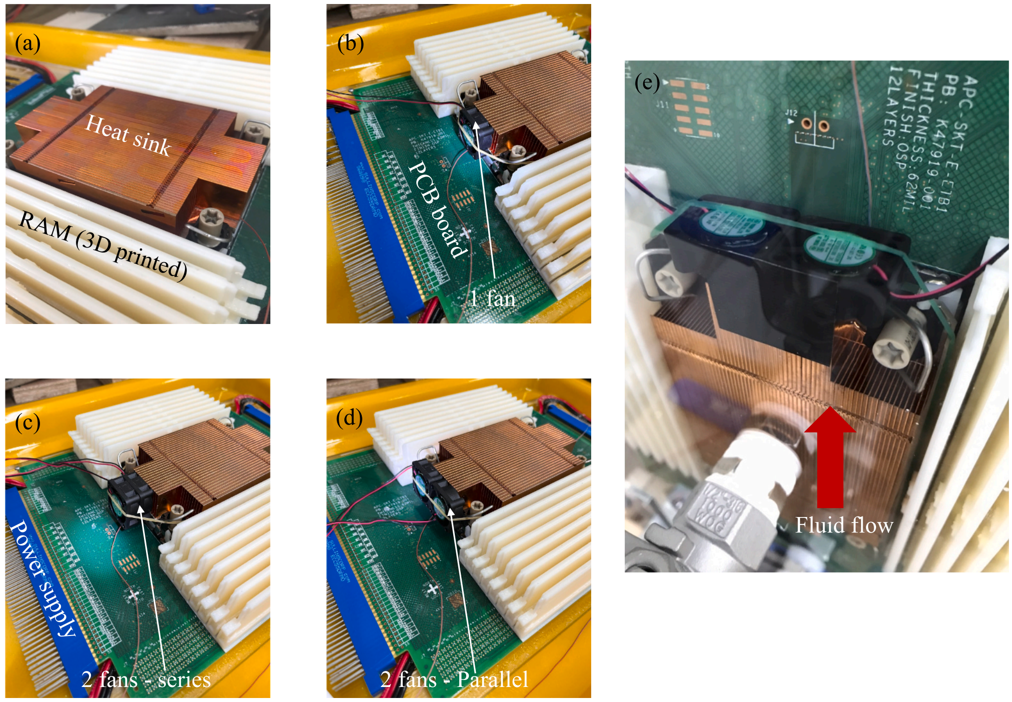

Fig. 12. Different fan arrangements downstream of the heat sink, (a) No fan, (b) Single fan, (c) Two fans with series arrangement, (d) Two fans with parallel arrangement, (e) picture showing the sever with 2 fans – parallel arrangement submerged in the 1U tank (fluid flow direction is marked to indicate that the fans are arranged downstream of the heat sink).

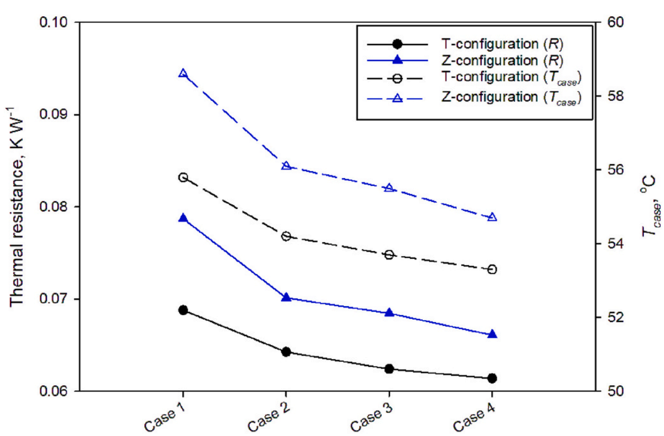

Fig. 13. Performance comparison of heat sinks with different fan arrangement. The X-axis refers to the different cases listed in Table 6. (Fin pitch = 1.2 mm, Flowrate = 3 LPM, $T_{fluid,in}$ = 35°C and Q_{in} = 300 W).

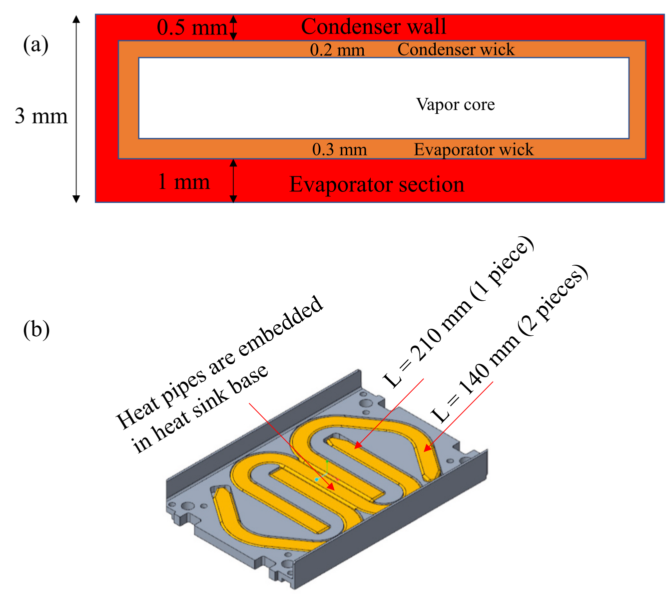

Fig. 14. (a) Details of the vapor chamber base and (b) Details of the heat pipe embedded heat sink base.

sink base are shown in Fig. 14(b). Three heat pipes (one heat pipe with a length of 210 mm and two heat pipes with a length of 140 mm) are embedded in the heat sink base. The total thickness of the heat pipe is 3 mm and its wall thickness is about 0.3 mm. Sintered copper powder with a size of 70 μm is used to produce the wick structure of the heat pipe.

The performance of the heat sink with solid plate, heat pipe, and vapor chamber bases is compared in Fig. 15. As compared to the flat plate base, the heat sink with vapor chamber base showed the best performance, followed by the heat pipe base. The heat sink with heat pipe and vapor chamber bases offered approximately 13% and 19% reduction in thermal resistance and 3.3 °C and 4.8 °C drop in case temperature over the heat sink with solid plate base, respectively. The vapor chamber or embedded heat pipe increases the effective thermal conductivity of the heat sink base with effective heat spreading, and acts as a near isothermal plate (less temperature non-uniformity compared to the flat plate base) due to the phase change phenomenon. As a result, the heat sink with vapor chamber and heat pipe base offers a lower thermal resistance over the conventional heat sink with solid plate base.

4. Conclusions

The present study experimentally investigates the performance of single-phase liquid immersion cooling systems applicable for a 1U data server using FC-40 dielectric fluid. The study considers the influence of different design and operating conditions, including inlet/outlet port

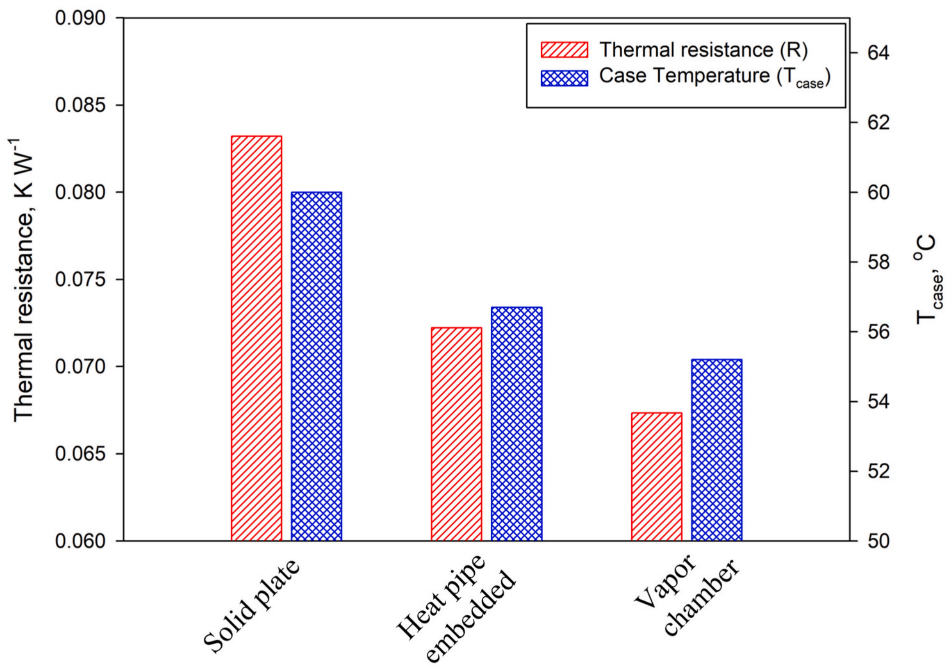

Fig. 15. Performance comparison of heat sinks with different bases such as flap plate, heat pipe, and vapor chamber. (T-configuration, Flowrate = 3 LPM, $T_{fluid,in}$ = 35^{o}C and Q_{in} = 300 W).

configuration (T and Z configuration), bypass effect (0–5 mm), flowrate (1–3 LPM), heating load (up to 600 W), fluid inlet temperature (15–35 °C), suction fan effect, and different heat sink base (solid plate, vapor chamber and heat pipe). The following conclusions are drawn based on the experimental analysis.

- The inlet/outlet ports arranged in a T-configuration outperformed the *Z*-configuration. The T-configuration offered about 12.6% and 0.5–2.8 °C reduction in thermal resistance and case temperature compared to those of the Z-configuration, respectively.
- The thermal resistance and case temperature can increase up to 7.4% and 1.5 °C when there is a gap of 5 mm (i.e., bypass effect), respectively.
- The thermal resistance is decreased with increasing flowrate, heating load, and inlet fluid temperature. An inlet temperature of 35 °C can offer about 8–13% reduction in thermal resistance when compared to the 15 °C.
- Among the different fan arrangements (no fan, single fan, two fans with serial arrangement, and two fans with parallel arrangement), a maximum of 11% and 2.5 °C reduction in thermal resistance and case temperature is obtained through two fans with parallel arrangement, respectively.
- Compared to the conventional heat sink with a solid plate base, the heat sink with heat pipe and vapor chamber bases offered a reduction of approximately 13% and 19% in thermal resistance and 3.3 °C and 4.8 °C drop in case temperature, respectively.

Authorship statement

M Muneeshwaran: Conceptualization, investigation, data curation, and writing-original draft.

Yueh-Cheng Lin: Conceptualization, experimentation.

Chi-Chuan Wang*: Conceptualization, supervision, reviewing and revising.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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